IOE AND AI DRIVEN SYSTEMS

EXPERIMENT 6 :

PROGRAM CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
# Set random seed for reproducibility
np.random.seed(42)
print("="*85)
print("PART 1: PEOPLE-PROCESS-DATA-THINGS (P-P-D-T) IoE & EDGE/FOG/CLOUD SIMULATION (WITH MQTT)")
print("="*85)
# 1. P-P-D-T Data Flow Simulation across Tiers (Edge, Fog, Cloud) including MQTT messaging
ppdt_components = [
    "Things (Sensors/Actuators + MQTT)",
    "People (User Wearables/UI + MQTT)",
    "Process (Node-RED/Rules)",
    "Data (Aggregated Store)"
]
tiers = ["Edge Layer", "Fog Layer", "Cloud Layer"]
ppdt_results = []
for comp in ppdt_components:
    for tier in tiers:
        # Simulate realistic latency, bandwidth usage, and packet loss based on tier proximity & MQTT behavior
        if tier == "Edge Layer":
            latency = np.random.uniform(2.0, 5.0)
            throughput = np.random.uniform(50.0, 100.0)
            reliability = np.random.uniform(99.0, 99.9)
```

```
        elif tier == "Fog Layer":
            latency = np.random.uniform(10.0, 25.0)
            throughput = np.random.uniform(200.0, 500.0)
            reliability = np.random.uniform(98.0, 99.5)
        else:  # Cloud Layer
            latency = np.random.uniform(50.0, 120.0)
            throughput = np.random.uniform(1000.0, 2500.0)
            reliability = np.random.uniform(97.0, 99.0)
            ppdt_results.append([comp, tier, round(latency, 2), round(throughput, 2), round(reliability, 2)])
df_ppdt = pd.DataFrame(ppdt_results, columns=["IoE Component (P-P-D-T + MQTT)", "Pipeline Tier", "Avg Latency (ms)", "Throughput (kbps)", "Reliability (%)"])
print(df_ppdt.head(12).to_string(index=False))
print("\n" + "="*85)
print("PART 2: 5G vs 6G COMMUNICATION BENCHMARKING FOR IoE")
print("="*85)
# 2. 5G vs 6G Benchmarking
technologies = ["5G Ultra-Reliable Low-Latency (URLLC)", "6G Terahertz (THz) Communication"]
benchmark_data = {
    "5G URLLC": {"Throughput": 10000, "Latency": 4.5, "Reliability": 99.99, "Energy": 12.0},
    "6G THz":   {"Throughput": 100000, "Latency": 0.5, "Reliability": 99.999, "Energy": 4.5}
}
bench_results = []
for tech in technologies:
    key = "5G URLLC" if "5G" in tech else "6G THz"
    b = benchmark_data[key]
    # Add small statistical variations
```

```
    throughput = max(100, b["Throughput"] + np.random.normal(0, b["Throughput"] *
0.02))

    latency = max(0.1, b["Latency"] + np.random.normal(0, 0.1))

    reliability = min(100.0, b["Reliability"] + np.random.normal(0, 0.001))

    energy = max(0.5, b["Energy"] + np.random.normal(0, 0.5))

     bench_results.append([tech, round(throughput, 2), round(latency, 2),
round(reliability, 3), round(energy, 2)])

df_bench = pd.DataFrame(bench_results, columns=["Communication Tech",
"Throughput (Mbps)", "Latency (ms)", "Reliability (%)", "Energy Consumption
(mJ/bit)"])

print(df_bench.to_string(index=False))

print("\n" + "="*85)

print("GENERATING BENCHMARKING & PIPELINE GRAPHS...")

print("="*85)

# 3. Visualizations

# Graph 1: Edge/Fog/Cloud Latency across P-P-D-T components (including MQTT)

plt.figure(figsize=(11, 5))

for tier in tiers:

    subset = df_ppdt[df_ppdt["Pipeline Tier"] == tier]

    plt.plot(subset["IoE Component (P-P-D-T + MQTT)"], subset["Avg Latency (ms)"],
marker='o', linewidth=2, label=tier)

plt.title("Edge / Fog / Cloud Pipeline Latency for P-P-D-T & MQTT Components")

plt.xlabel("IoE Components (P-P-D-T + MQTT)")

plt.ylabel("Latency (ms)")

plt.xticks(rotation=10)

plt.legend()

plt.grid(True, alpha=0.3)

plt.tight_layout()

plt.show()


# Graph 2: 5G vs 6G Comparison Bar Charts
```

```
metrics = [
    ("Throughput (Mbps)", "Throughput Comparison: 5G vs 6G", "Throughput (Mbps)"),
    ("Latency (ms)", "Latency Comparison: 5G vs 6G", "Latency (ms)"),
    ("Reliability (%)", "Reliability Comparison: 5G vs 6G", "Reliability (%)"),
    ("Energy Consumption (mJ/bit)", "Energy Efficiency Comparison: 5G vs 6G", "Energy (mJ/bit)")
]
for col, title, ylabel in metrics:
    plt.figure(figsize=(7, 4))
    bars = plt.bar(df_bench["Communication Tech"], df_bench[col], color=["#3498db", "#2ecc71"])
    plt.title(title)
    plt.ylabel(ylabel)
    plt.grid(axis="y", alpha=0.3)
    plt.tight_layout()
    plt.show()
print("\n" + "="*85)
print("CONCLUSION & SUMMARY")
print("="*85)
print("1. Edge/Fog pipeline integrated with lightweight MQTT messaging minimizes local processing delay for Things and People components.")
print("2. Cloud layer handles heavier aggregated P-P-D-T data storage with higher latency.")
print("3. 6G significantly outperforms 5G in throughput, ultra-low latency (<1ms), and energy efficiency,")
print("   making it optimal for massive next-generation IoE deployments.")
```

OUTPUT:

PART 1: PEOPLE-PROCESS-DATA-THINGS (P-P-D-T) IoE & EDGE/FOG/CLOUD SIMULATION (WITH MQTT)

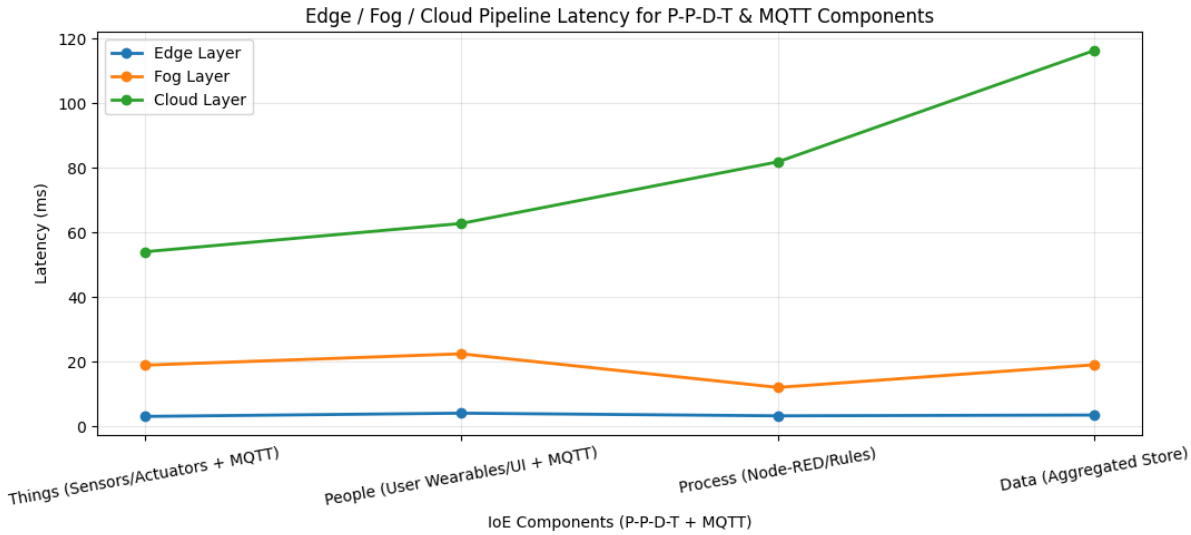
IoE Component (P-P-D-T + MQTT)	Pipeline Tier	Avg Latency (ms)	Throughput (kbps)	Reliability (%)
Things (Sensors/Actuators + MQTT)	Edge Layer	3.12	97.54	99.66
Things (Sensors/Actuators + MQTT)	Fog Layer	18.98	246.81	98.23
Things (Sensors/Actuators + MQTT)	Cloud Layer	54.07	2299.26	98.20
People (User Wearables/UI + MQTT)	Edge Layer	4.12	51.03	99.87
People (User Wearables/UI + MQTT)	Fog Layer	22.49	263.70	98.27
People (User Wearables/UI + MQTT)	Cloud Layer	62.84	1456.36	98.05
Process (Node-RED/Rules)	Edge Layer	3.30	64.56	99.55
Process (Node-RED/Rules)	Fog Layer	12.09	287.64	98.55
Process (Node-RED/Rules)	Cloud Layer	81.92	2177.76	97.40
Data (Aggregated Store)	Edge Layer	3.54	79.62	99.04
Data (Aggregated Store)	Fog Layer	19.11	251.16	98.10
Data (Aggregated Store)	Cloud Layer	116.42	2448.45	98.62

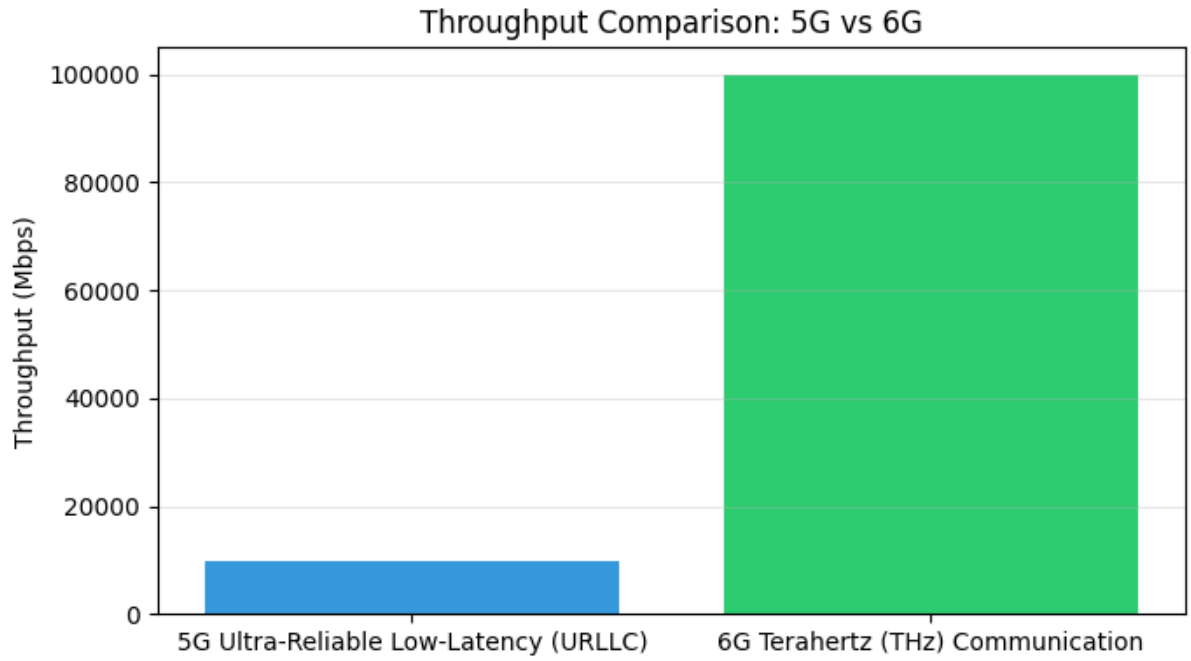
PART 2: 5G vs 6G COMMUNICATION BENCHMARKING FOR IoE

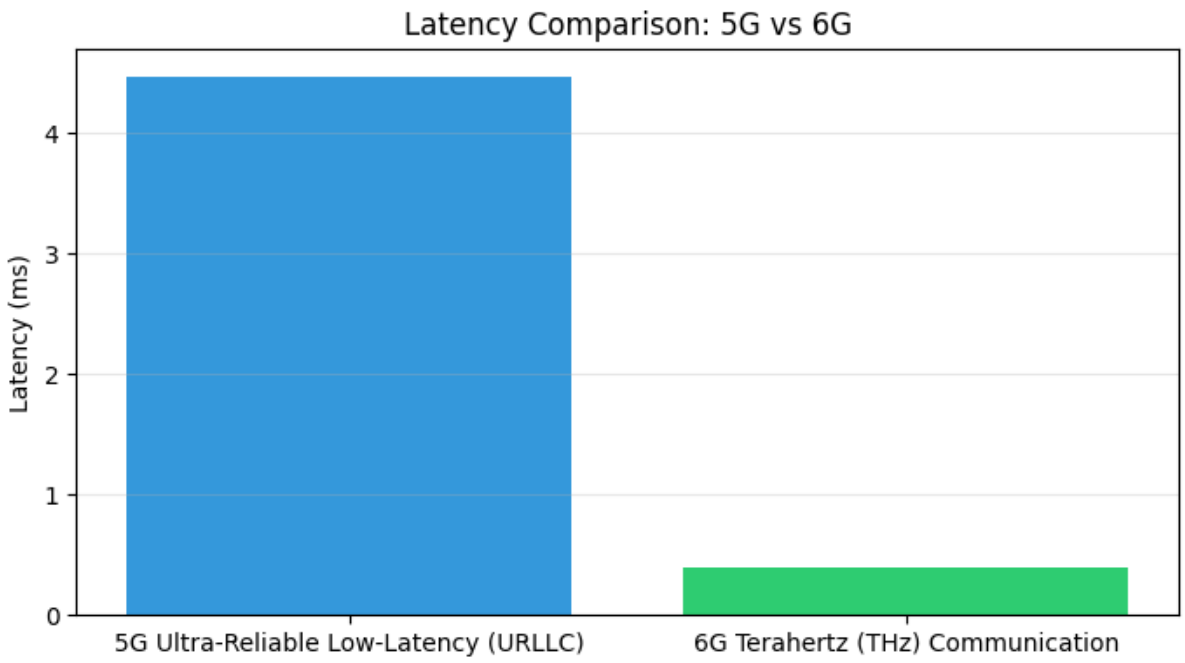
Communication Tech	Throughput (Mbps)	Latency (ms)	Reliability (%)	Energy Consumption (mJ/bit)
5G Ultra-Reliable Low-Latency (URLLC)	9879.87	4.47	99.989	12.93

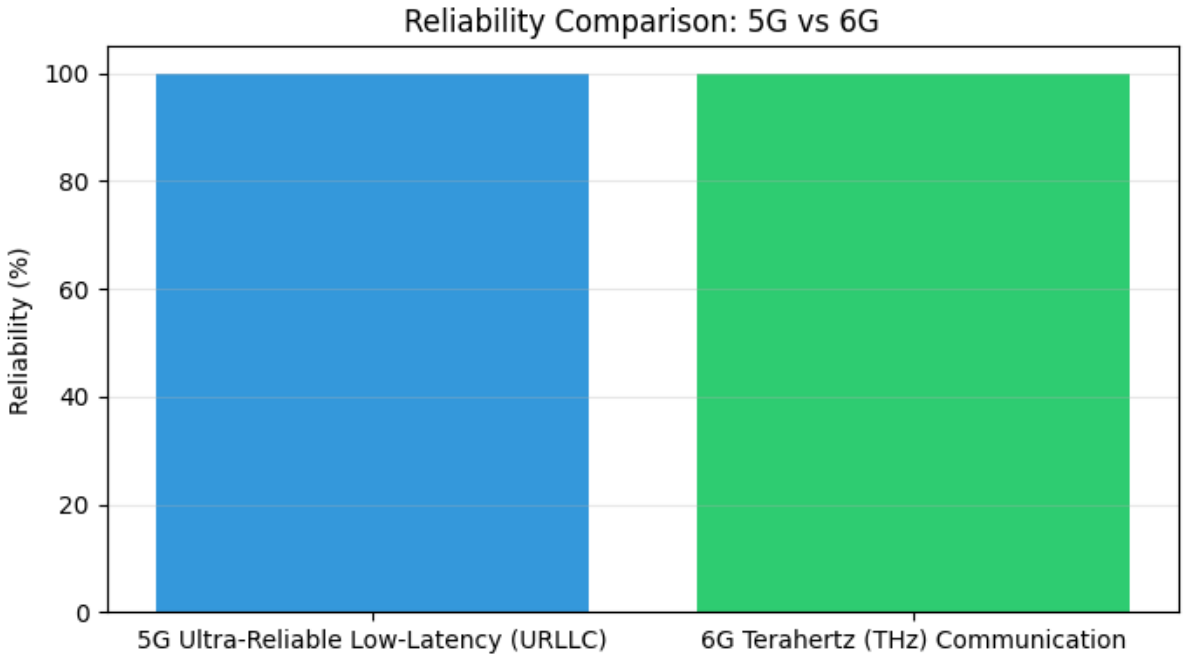
6G Terahertz (THz) Communication 99973.01 0.39 100.000 3.89

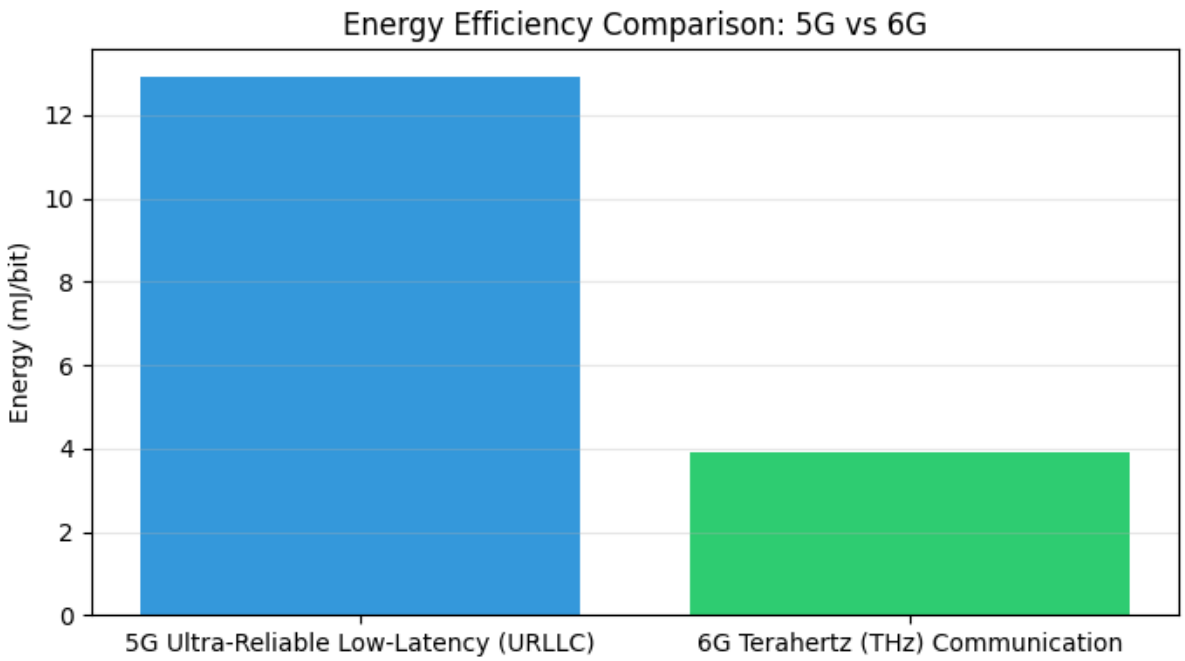
GENERATING BENCHMARKING & PIPELINE GRAPHS...






Latency Comparison: 5G vs 6G
Latency (ms)
0
1
2
3
4
5G Ultra-Reliable Low-Latency (URLLC)
6G Terahertz (THz) Communication


Reliability Comparison: 5G vs 6G
Reliability (%)
0
20
40
60
80
100
5G Ultra-Reliable Low-Latency (URLLC)
6G Terahertz (THz) Communication



CONCLUSION & SUMMARY

1. Edge/Fog pipeline integrated with lightweight MQTT messaging minimizes local processing delay for Things and People components.

2. Cloud layer handles heavier aggregated P-P-D-T data storage with higher latency.

3. 6G significantly outperforms 5G in throughput, ultra-low latency (<1ms), and energy efficiency,

making it optimal for massive next-generation IoE deployments.